

Freestanding and Flexible Micrometer-Thick PEDOT:PSS Film with High Power Factor

Liwei Shi, Sooyon Chang, and Zhiting Tian*

Cite This: *ACS Appl. Energy Mater.* 2025, 8, 25–30

Read Online

ACCESS |



Metrics & More



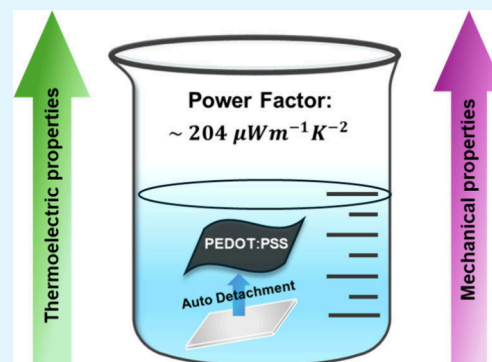
Article Recommendations



Supporting Information

ABSTRACT: High-efficiency thermoelectric materials are vital for sustainable energy. This work presents a reliable method to fabricate freestanding, flexible micrometer-thick films of poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS), achieving a power factor of $204 \mu\text{W m}^{-1} \text{K}^{-2}$, 4 orders of magnitude higher than pristine films. This improvement stems from polymer chain realignment and enhanced charge transport via solvent doping, coupled with PSS reduction during post-treatment, which also strengthens film mechanics. Crucially, the freestanding nature ensures transferability, making them ideal for versatile applications. Besides thermoelectrics, these highly conductive, robust, and flexible PEDOT:PSS films provide attractive solutions for organic solar cells, battery electrodes, and flexible electronics.

KEYWORDS: Organic Thermoelectrics, PEDOT:PSS, High Power Factor, Freestanding, Highly Conductive, Enhanced Mechanical Properties, Doping, Post-treatment



Poly(3,4-ethylenedioxythiophene):poly(styrenesulfonate) (PEDOT:PSS) is a conducting polymer that has attracted considerable interest due to its unique properties, such as flexibility, optical transparency, and ease of processing, which make it a promising material in various fields. Nevertheless, the application of pristine PEDOT:PSS in organic thermoelectric generators (TEGs) has been constrained by its low thermoelectric performance. The thermoelectric performance of a material is determined by the dimensionless, temperature-dependent figure of merit ZT , which depends on the Seebeck coefficient (S), electrical conductivity (σ), and thermal conductivity (k).¹ Pristine PEDOT:PSS exhibits a very low power factor ($\text{PF} = S^2\sigma$), leading to poor thermoelectric performance compared to other conductive materials.

In recent years, extensive efforts have focused on enhancing the thermoelectric performance of PEDOT:PSS through various additives, including graphene, carbon nanotubes, or ionic liquids, or combined with some organic solvents, acids, and bases. These modifications primarily aim to increase charge carrier concentration, induce conformational rearrangements in PEDOT and PSS chains, and shift the Fermi level, collectively leading to improved charge transport and power factor.^{2–5} Most studies have utilized spin-coated PEDOT:PSS films, known for their high surface wettability and uniformity. For example, mixing PEDOT:PSS dispersion with ionic liquid 1-ethyl-3-methylimidazolium tricyanomethanide (EMIM:TCM) resulted in a film with improved PF of $175 \mu\text{W m}^{-1} \text{K}^{-2}$,⁶ treating PEDOT:PSS with dimethyl sulfoxide (DMSO) and hydrazine led to a PF of $318.4 \mu\text{W m}^{-1} \text{K}^{-2}$.⁷

Despite the high power factor, these techniques generally produce nanometer-thick films tightly bound to substrates, limiting practical use. Micrometer-thick PEDOT:PSS films were fabricated for enhanced mechanical stability and application viability. For example, PEDOT:PSS combined with carbon nanotube (CNT) and processed via vacuum filtration achieved a PF of $1.9 \mu\text{W m}^{-1} \text{K}^{-2}$.⁸ Formic acid and hydrazine treatment of drop-cast PEDOT:PSS films raised the PF to $93.5 \mu\text{W m}^{-1} \text{K}^{-2}$.² These films are, however, rarely suspended from substrates and often lack high thermoelectric properties due to challenges in achieving uniformity and precise structural control. Freestanding films offer advantages like transferability and easier handling and processing, yet methods to produce them remain underdeveloped.

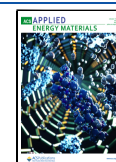
In this study, we present an effective and efficient method for producing freestanding micrometer-thick films of PEDOT:PSS that demonstrates a high power factor with excellent repeatability. Our method involves organic solvent doping, a straightforward and scalable drop-casting process followed by a thermal annealing step, and post-treatment (Supporting Information Figure S1). These freestanding films enable

Received: October 9, 2024

Revised: December 16, 2024

Accepted: December 17, 2024

Published: December 18, 2024



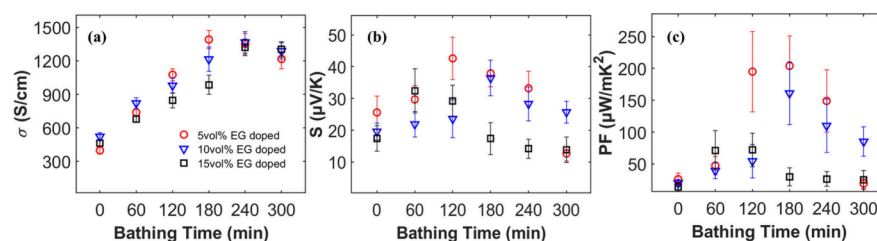


Figure 1. (a) EG-doped PEDOT:PSS films' electrical conductivity, (b) EG-doped PEDOT:PSS films' Seebeck coefficient, and (c) EG-doped PEDOT:PSS films' thermoelectric power factor at $T = 298$ K with different doping levels during the EG bathing post-treatment process.

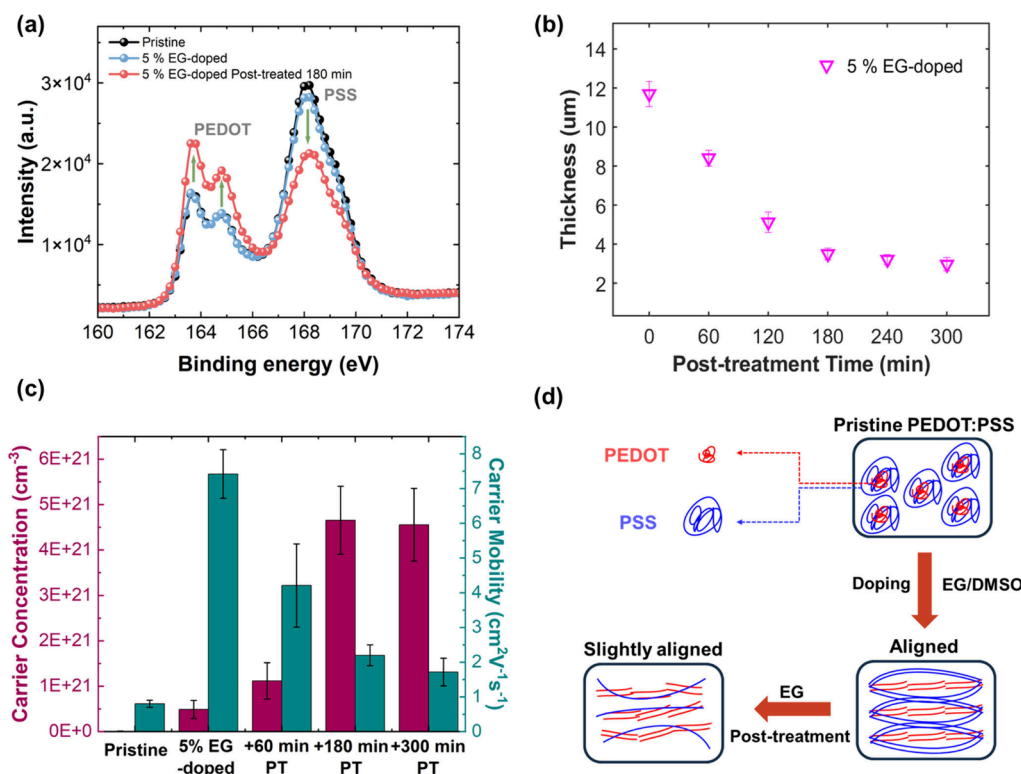


Figure 2. (a) XPS spectra (S2p) scan) of pristine PEDOT:PSS, 5 vol % EG-doped PEDOT:PSS, and 5 vol % EG-doped with additional 180 min post-treatment PEDOT:PSS micrometer-thick films. (b) Thickness measurements of 5 vol % EG-doped PEDOT:PSS with additional 0–300 min post-treatment. (c) Carrier concentration and mobility of PEDOT:PSS micrometer-thick films including pristine films, 5 vol % EG-doped films, and 5 vol % EG-doped with an additional 0–300 min post-treatment films. (d) Schematic representation of the structural reorientation of the PEDOT:PSS films by solvent doping and post-treatment.

effortless transfer across substrates, broadening their applicability. Notably, these films showcase a remarkable PF of up to $204 \mu\text{W m}^{-1} \text{K}^{-2}$, demonstrating that freestanding micrometer-thick films can deliver superior thermoelectric performance. The greater thickness also provides enhanced mechanical resilience without sacrificing flexibility, suitable for more durable thermoelectric applications. Furthermore, these films possess high electrical conductivity, which is ideal for flexible electronics and battery electrodes.

Figure 1 displays the thermoelectric properties (σ , S , and PF) of doped PEDOT:PSS micrometer-thick films following EG bath post-treatment. Measurements were consistently taken from the center region of the films, where the uniformity is much better than the edge. Pristine PEDOT:PSS films show low σ (0.455 S cm^{-1}) and S ($16.3 \mu\text{V K}^{-1}$). Polar solvent dopants like ethylene glycol (EG) and DMSO have been widely used to improve the thermoelectric properties by modifying polymer chain morphology.⁹ After doping with EG, σ increased by 3 orders of magnitude, reaching 521 S cm^{-1} (10

vol % EG). S also observed a modest improvement, reaching $25.6 \mu\text{V K}^{-1}$ (5 vol % EG). To further improve the PF of PEDOT:PSS micrometer-thick films, a bath method of post-treatment was carried out instead of the standard drop method because it lacks sufficient penetration for thicker films. Here, EG-doped films were immersed in an EG solution for a specified time, ensuring more uniform distribution and better interaction. Both σ and S of the films were further improved. The maximum σ reached 1392 S cm^{-1} after 3 h of post-treatment with a 5 vol % doping level, while the peak S attained $42.6 \mu\text{V K}^{-1}$ after 2 h with the same doping level. The highest PF achieved was $204 \mu\text{W K}^{-2} \text{ m}^{-1}$ for 5 vol % EG-doped PEDOT:PSS films that underwent 3 h of EG bath post-treatment, outperforming other reported micrometer-thick PEDOT:PSS films.^{2,8} We have also carried out solvent doping using another dopant, DMSO, and found that EG was more effective in modifying the polymer chain morphology, leading to better properties (Figure S3).

We conducted various characterizations to investigate the mechanism of solvent doping and EG bath post-treatment on thermoelectric (TE) properties enhancement of PEDOT:PSS micrometer-thick films. Figure 2a presents the X-ray photo-emission spectroscopy (XPS) spectra of pristine PEDOT:PSS films, 5 vol % EG-doped PEDOT:PSS films, and EG-doped PEDOT:PSS films after an additional 3 h of post-treatment, yielding the highest PF. XPS analysis confirms and quantifies the reduction of PSS following EG bath post-treatment. As depicted in Figure 2a, the S(2p) peak within 167–170 eV represents sulfur in PSS, while the double peaks within 163–166 eV represent sulfur in PEDOT.³ EG-doped films exhibit nearly identical S(2p) peak intensities for both PEDOT and PSS as pristine films, suggesting the solvent doping process does not contribute to the reduction of PSS. However, EG bath post-treatment notably decreases PSS S(2p) intensity and increases PEDOT intensity, indicating a higher proportion of PEDOT due to the substantial removal of PSS segments, which reduces film thickness, as shown in Figure 2b.

Hall effect measurements (Figure 2c) were conducted on pristine PEDOT:PSS films and 5 vol % EG-doped films with or without post-treatment to assess changes in carrier concentration (n) and mobility (μ), which directly impact σ and S . While Hall measurements may not be precise for quantitatively capturing the mobility change with carrier concentration in disordered organic systems, they still offer valuable qualitative insights into the connection between mobility and the internal structure. Solvent doping significantly increased both n and μ , with pristine films showing an n of $7.62 \times 10^{18} \text{ cm}^{-3}$ and μ of $0.81 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$, raising to $3.48 \times 10^{20} \text{ cm}^{-3}$ and $7.42 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$ in 5 vol % EG-doped films. In pristine PEDOT:PSS, PEDOT-rich cores are usually surrounded by PSS-rich shells, creating an entangled core-shell structure.¹⁰ EG or DMSO doping induces phase separation between PEDOT and PSS and facilitates the realignment of intergrain chains, forming a more ordered linear structure (Figure 2d).¹¹ This structural ordering boosts the number of polarons or bipolarons (charge carriers) along the monomer units, enhancing both carrier concentration and mobility.^{12,13} EG bath post-treatment further increases n but reduces μ with prolonged treatment due to insulated PSS removal, which shrinks the volume and raises the relative percentage of conductive PEDOT.^{14,15} Despite this, PSS is crucial for spacing, solubilizing PEDOT segments, and providing sulfonate counterions that stabilize positive charges on the PEDOT backbone, aiding charge carrier mobility.^{16,17} Hence, removing PSS also causes reclustering of PEDOT grains, resulting in lower μ .¹⁸ According to Figure 2c, the maximum n ($4.66 \times 10^{21} \text{ cm}^{-3}$) was observed in 5 vol % EG-doped PEDOT:PSS films after 3 h of post-treatment, with minimal thickness reduction, indicating near-complete PSS removal. While there was no further noticeable increase in n , the ongoing aggregation of PEDOT grains continued to lower μ , resulting in the drop of σ . This suggests that 5 vol % EG doping with a 3 h EG bath post-treatment is the optimal way to reach a high PF.

Raman spectroscopy (Figure 3a,b) was employed to delve into the enhanced thermoelectric (TE) properties of PEDOT:PSS micrometer-thick films, with characteristic peaks identified between 1200 and 1600 cm^{-1} . In pristine PEDOT:PSS films, peaks observed at around 1265 and 1433 cm^{-1} correspond to the C_α – C_α inter-ring stretching vibration and the C_α – C_β symmetric stretching vibration, respectively.^{19,20} The band at approximately 1572 cm^{-1} corresponds

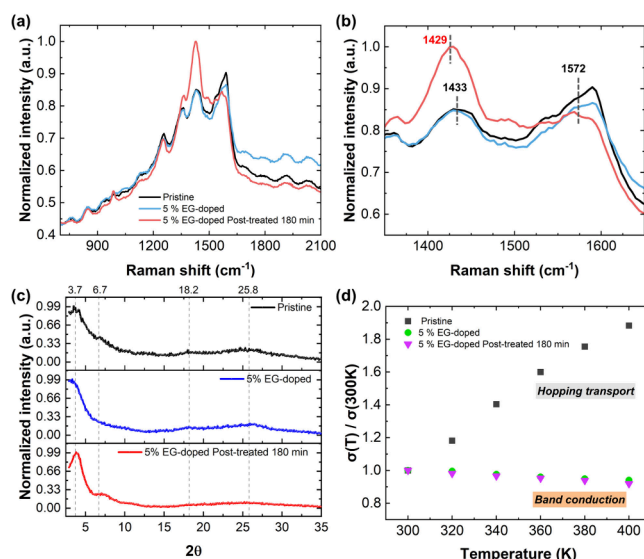


Figure 3. (a) Raman spectra from 700 to 2100 cm^{-1} . (b) Raman spectra from 1350 to 1650 cm^{-1} . (c) 2θ of in-plane diffraction XRD shows that packing of PEDOT:PSS changes with the removal of PSS. (d) Temperature dependent electrical conductivity (normalized to 300 K) of pristine PEDOT:PSS, 5 vol % EG-doped PEDOT:PSS, and 5 vol % EG-doped with additional 180 min post-treatment PEDOT:PSS thick films from 300 to 400 K.

to a doping-induced peak emerging from the split in the asymmetrical C_α – C_β stretching mode band, indicative of the oxidized state of PEDOT:PSS.²¹ Within that band range, the intensity slightly decreased relative to pristine films after doping with EG, suggesting structural alterations such as phase separation and PEDOT chains elongation that enhance intergrain interactions, thereby increasing both the concentration and mobility of carriers, as confirmed by Hall effect measurements.^{19,22} Following an additional 180 min of post-treatment, the prominent band shifted from 1433 to 1429 cm^{-1} and narrowed, reflecting a red shift to lower energy vibration modes. This shift suggests a resonant structure transition of PEDOT chains from benzenoid structure into quinoid structure, which is more favorable for intragrain (intra- and interchain) charge transport in PEDOT.^{23–25} This conversion mitigates the negative impact of the intergrain mobility reduction caused by the loss of the stabilizer PSS, which is crucial for the further enhancement of electrical conductivity after post-treatment.

XRD measurements (Figure 3c) were performed using an in-plane diffraction geometry, and the lattice spacing was calculated from the observed Bragg reflection peaks. The XRD patterns of pristine PEDOT:PSS films exhibit characteristic Bragg reflections at 2θ of 3.7°, 6.7°, 18.2°, and 25.8°. The peaks at 3.7° and 6.7° correspond to the (100) low-angle reflection arising from the lamellar stacking of PEDOT chains. After the post-treatment, the peak at $2\theta = 3.7^\circ$ intensified, suggesting an improvement in the crystallinity and abundance of PEDOT chains (Figure S4). These PEDOT chains were observed through SEM (Figure S5), where the 180 min post-treated sample had a small interlamellar distance and higher crystallinity compared to the pristine and 5% EG-doped samples. Additionally, a higher angle shift of this peak indicates a more condensed packing resulting from the removal of PSS also observable through the disappearance of the peak at $2\theta = 18.2^\circ$, which corresponds to the weak π – π stacking of PSS.

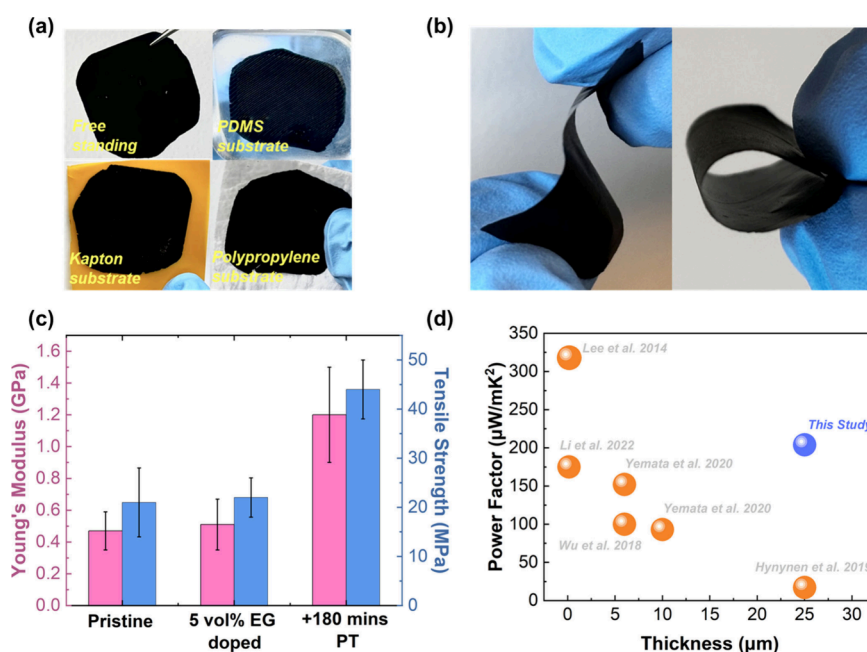


Figure 4. (a) Freestanding micrometer-thick PEDOT:PSS films transferred to different substrates. (b) Flexible and bendable freestanding micrometer-thick films. (c) Young's modulus and tensile strength of pristine PEDOT:PSS, 5 vol % EG-doped PEDOT:PSS, and 5 vol % EG-doped with additional 180 min post-treatment PEDOT:PSS micrometer-thick films. (d) Comparison of PEDOT:PSS films in this work with other reported organic thermoelectric films in terms of thickness and power factor.^{2,6,7,28–30}

The peaks at $2\theta = 25.8^\circ$, representing the strong π - π stacking of thiophene rings in PEDOT, exhibited a shift toward higher angles after doping and post-treatment.²² This shift suggests a reduction in the interchain distances, enabling more efficient charge transfer through short-range interactions and enhancing the overall electrical conductivity of PEDOT:PSS films.

To further understand the charge transport behavior of PEDOT:PSS micrometer-thick films, temperature-dependent electrical conductivity measurement was conducted from 50 to 400 K (Figure S6). Focusing on ambient temperatures, Figure 3d shows data from 300 to 400 K, revealing two distinct conductivity regimes with specific temperature dependencies and electronic properties. Pristine PEDOT:PSS films demonstrate semiconducting behavior, with a positive temperature coefficient, indicating thermally activated charge transport via carrier hopping between localized states.²⁶ In PEDOT:PSS, the Fermi level lies among localized states in the middle between the valence band and bipolaron band.⁵ Doping PEDOT:PSS films with EG reduces chain disorder, resulting in bipolaron band delocalization and correspondingly metallic transport, marked by a negative temperature coefficient.⁴ Due to the empty spin in the bipolaron band and asymmetry in the density of states around the Fermi level, both σ and S increase.⁵

Compared to nanofilms of PEDOT:PSS, micrometer-thick films exhibit notably superior mechanical properties. Our freestanding micrometer-thick films can be transferred to various substrates after EG bath post-treatment and securely adhered among drying (Figure 4a). Additionally, these films are fully flexible and foldable, as shown in Figure 4b. A tensile test was performed under ambient conditions to evaluate their mechanical robustness and reveal the impact of EG doping and subsequent post-treatment on their physical integrity (Figure S7). This test involved stretching micrometer-thick films of pristine PEDOT:PSS, PEDOT:PSS doped with 5 vol % EG, and PEDOT:PSS films doped with 5 vol % EG followed with

an additional 180 min of post-treatment until breakage. The dimensions of these samples are detailed in Figure S7a. Figure 4c presents tensile test results, including Young's modulus and tensile strength for the examined PEDOT:PSS films. The observed values for both parameters (approximately 0.5 GPa for Young's modulus and about 22 MPa for tensile strength) were remarkably consistent between pristine films and EG-doped films, indicating that solvent doping does not compromise the films' mechanical integrity despite microscopic changes in polymer configuration. However, after 180 min of post-treatment in the EG bath, a significant enhancement was observed in both Young's modulus and the tensile strength of the material, which escalated to approximately 1.2 GPa and 44 MPa, respectively. This improvement indicates that the PEDOT:PSS micrometer-thick films underwent a transformation after removing a large amount of PSS, becoming stiffer and capable of withstanding greater stress levels. Modified PEDOT:PSS exhibits reduced affinity toward moisture in the surrounding air.²⁷ This alteration in hydrophobicity, together with the higher crystallinity of PEDOT lamellae, leads to an enhancement in the mechanical properties of the material. Thermogravimetric analysis (TGA) was also conducted on PEDOT:PSS micrometer-thick films with optimal PF, as shown in Figure S8. The result indicates that these films can operate stably at a temperature of 200 °C, making them highly practical for various applications. With enhanced mechanical properties from increased thickness, together with a high PF, our approach offers significant overall advantages over other films in the field of organic thermoelectric. (Figure 4d).^{2,6,7,28–30}

In summary, our study introduces an innovative and straightforward method for fabricating freestanding micrometer-thick PEDOT:PSS films with outstanding power factor. Using a combination of organic solvent doping, drop-casting, thermal annealing, and post-treatment, we developed PEDOT:PSS thick films that not only overcome the limitations

associated with traditional nanoscale thin films but also feature enhanced mechanical stability, ease of handling, and transferability. Notably, processed films exhibit an optimal power factor of up to $204 \mu\text{W m}^{-1} \text{K}^{-2}$, significantly higher than other existing micrometer-thick films. The improvement primarily stems from alterations in alignment and charge transport mechanisms induced by solvent doping as well as the reduction of PSS content during the post-treatment process, which is feasible through the introduction of a single polar solvent, EG. The bath method for post-treatment provides greater efficiency than the drop method due to increased contact area and optimized bathing time. Additionally, PSS reduction also significantly contributes to the bolstering of the films' mechanical properties. This work therefore provides a scalable and effective pathway to explore the full potential of PEDOT:PSS for energy harvesting and storage as well as flexible electronics applications.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsaem.4c02568>.

Experimental section, additional experimental results, and further material characterization including XRD data, SEM images, full range temperature-dependent electrical conductivity plot, tensile test data, and TGA data (PDF)

■ AUTHOR INFORMATION

Corresponding Author

Zhiting Tian – Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca, New York 14853, United States; orcid.org/0000-0002-5098-7507; Email: zhiting@cornell.edu

Authors

Liwei Shi – Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca, New York 14853, United States; orcid.org/0009-0004-6517-9782

Sooyon Chang – Sibley School of Mechanical and Aerospace Engineering, Cornell University, Ithaca, New York 14853, United States; orcid.org/0009-0000-4099-2313

Complete contact information is available at: <https://pubs.acs.org/doi/10.1021/acsaem.4c02568>

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

The authors gratefully acknowledge the support from DOE (Grant No. DE-EE0009771). This work made use of the Cornell Center for Materials Research Shared Facilities that are supported through the NSF MRSEC program (Grant No. DMR-1719875).

■ REFERENCES

- (1) Li, C.; Ma, H.; Tian, Z. Thermoelectric Properties of Crystalline and Amorphous Polypyrrole: A Computational Study. *Appl. Therm. Eng.* **2017**, *111*, 1441–1447.
- (2) Yemata, T. A.; Zheng, Y.; Kyaw, A. K. K.; Wang, X.; Song, J.; Chin, W. S.; Xu, J. Modulation of the Doping Level of PEDOT:PSS Film by Treatment with Hydrazine to Improve the Seebeck Coefficient. *RSC Adv.* **2020**, *10* (3), 1786–1792.
- (3) Jin Bae, E.; Hun Kang, Y.; Jang, K.-S.; Yun Cho, S. Enhancement of Thermoelectric Properties of PEDOT:PSS and Tellurium-PEDOT:PSS Hybrid Composites by Simple Chemical Treatment. *Sci. Rep.* **2016**, *6* (1), No. 18805.
- (4) Yoon, C. O.; Reghu, M.; Moses, D.; Heeger, A. J.; Cao, Y.; Chen, T.-A.; Wu, X.; Rieke, R. D. Hopping Transport in Doped Conducting Polymers in the Insulating Regime near the Metal-Insulator Boundary: Polypyrrole, Polyaniline and Polyalkylthiophenes. *Synth. Met.* **1995**, *75* (3), 229–239.
- (5) Bubnova, O.; Khan, Z. U.; Wang, H.; Braun, S.; Evans, D. R.; Fabretto, M.; Hojati-Talemi, P.; Dagnelund, D.; Arlin, J.-B.; Geerts, Y. H.; Desbief, S.; Breiby, D. W.; Andreasen, J. W.; Lazzaroni, R.; Chen, W. M.; Zozoulenko, I.; Fahlman, M.; Murphy, P. J.; Berggren, M.; Crispin, X. Semi-Metallic Polymers. *Nat. Mater.* **2014**, *13* (2), 190–194.
- (6) Li, X.; Zou, R.; Liu, Z.; Mata, J.; Storer, B.; Chen, Y.; Qi, W.; Zhou, Z.; Zhang, P. Deciphering the Superior Thermoelectric Property of Post-Treatment-Free PEDOT:PSS/IL Hybrid by X-Ray and Neutron Scattering Characterization. *npj Flexible Electronics* **2022**, *6* (1), 6.
- (7) Lee, S. H.; Park, H.; Kim, S.; Son, W.; Cheong, I. W.; Kim, J. H. Transparent and Flexible Organic Semiconductor Nanofilms with Enhanced Thermoelectric Efficiency. *J. Mater. Chem. A* **2014**, *2* (20), 7288–7294.
- (8) He, X.; Shi, J.; Hao, Y.; Wang, L.; Qin, X.; Yu, J. PEDOT:PSS/CNT Composites Based Ultra-Stretchable Thermoelectrics and Their Application as Strain Sensors. *Compos. Commun.* **2021**, *27*, No. 100822.
- (9) Urban, J. J.; Menon, A. K.; Tian, Z.; Jain, A.; Hippalgaonkar, K. New Horizons in Thermoelectric Materials: Correlated Electrons, Organic Transport, Machine Learning, and More. *J. Appl. Phys.* **2019**, *125* (18), 180902.
- (10) Modarresi, M.; Zozoulenko, I. Why Does Solvent Treatment Increase the Conductivity of PEDOT: PSS? Insight from Molecular Dynamics Simulations. *Phys. Chem. Chem. Phys.* **2022**, *24* (36), 22073–22082.
- (11) Wei, Q.; Mukaida, M.; Naitoh, Y.; Ishida, T. Morphological Change and Mobility Enhancement in PEDOT:PSS by Adding Cosolvents. *Adv. Mater.* **2013**, *25* (20), 2831–2836.
- (12) Kumar, S. R. S.; Kurra, N.; Alshareef, H. N. Enhanced High Temperature Thermoelectric Response of Sulphuric Acid Treated Conducting Polymer Thin Films. *J. Mater. Chem. C Mater.* **2016**, *4* (1), 215–221.
- (13) Shahrim, N. A.; Ahmad, Z.; Wong Azman, A.; Fachmi Buys, Y.; Sarifuddin, N. Mechanisms for Doped PEDOT:PSS Electrical Conductivity Improvement. *Mater. Adv.* **2021**, *2* (22), 7118–7138.
- (14) Song, I.; Yeon Park, N.; Seung Jeong, G.; Hwan Kang, J.; Hwa Seo, J.; Choi, J.-Y. Conductive Channel Formation for Enhanced Electrical Conductivity of PEDOT:PSS with High Work-Function. *Appl. Surf. Sci.* **2020**, *529*, No. 147176.
- (15) Ko, Y.; Kim, J.; Kim, D.; Kwon, G.; Yamauchi, Y.; You, J. Fabrication of Highly Conductive Porous Cellulose/PEDOT:PSS Nanocomposite Paper via Post-Treatment. *Nanomaterials* **2019**, *9* (4), 612.
- (16) Sun, K.; Zhang, S.; Li, P.; Xia, Y.; Zhang, X.; Du, D.; Isikgor, F. H.; Ouyang, J. Review on Application of PEDOTs and PEDOT:PSS in Energy Conversion and Storage Devices. *J. Mater. Sci.: Mater. Electron.* **2015**, *26* (7), 4438–4462.
- (17) Ji, C.; Zhou, C.; Zhao, B.; Yang, J.; Zhao, J. Effect of Counterion Binding to Swelling of Polyelectrolyte Brushes. *Langmuir* **2021**, *37* (18), 5554–5562.
- (18) Dhar, S.; Majumder, T.; Chakraborty, P.; Mondal, S. P. DMSO Modified PEDOT:PSS Polymer/ZnO Nanorods Schottky Junction Ultraviolet Photodetector: Photoresponse, External Quantum Efficiency, Detectivity, and Responsivity Augmentation Using N Doped Graphene Quantum Dots. *Org. Electron.* **2018**, *53*, 101–110.
- (19) Park, H.; Lee, S. H.; Kim, F. S.; Choi, H. H.; Cheong, I. W.; Kim, J. H. Enhanced Thermoelectric Properties of PEDOT:PSS

Nanofilms by a Chemical Dedoping Process. *J. Mater. Chem. A* **2014**, 2 (18), 6532–6539.

(20) Chang, S. H.; Chiang, C.-H.; Kao, F.-S.; Tien, C.-L.; Wu, C.-G. Unraveling the Enhanced Electrical Conductivity of PEDOT:PSS Thin Films for ITO-Free Organic Photovoltaics. *IEEE Photonics J.* **2014**, 6 (4), 1–7.

(21) Xiong, S.; Zhang, L.; Lu, X. Conductivities Enhancement of Poly(3,4-Ethylenedioxythiophene)/Poly(Styrene Sulfonate) Transparent Electrodes with Diol Additives. *Polym. Bull.* **2013**, 70 (1), 237–247.

(22) Hosseini, E.; Ozhukil Kollath, V.; Karan, K. The Key Mechanism of Conductivity in PEDOT:PSS Thin Films Exposed by Anomalous Conduction Behaviour upon Solvent-Doping and Sulfuric Acid Post-Treatment. *J. Mater. Chem. C Mater.* **2020**, 8 (12), 3982–3990.

(23) Ouyang, J.; Xu, Q.; Chu, C.-W.; Yang, Y.; Li, G.; Shinar, J. On the Mechanism of Conductivity Enhancement in Poly(3,4-Ethylenedioxythiophene):Poly(Styrene Sulfonate) Film through Solvent Treatment. *Polymer (Guildf)* **2004**, 45 (25), 8443–8450.

(24) Wang, X.; Feng, G.; Ge, M. Influence of Ethylene Glycol Vapor Annealing on Structure and Property of Wet-Spun PVA/PEDOT:PSS Blend Fiber. *J. Mater. Sci.* **2017**, 52 (12), 6917–6927.

(25) Chou, T.-R.; Chen, S.-H.; Chiang, Y.-T.; Lin, Y.-T.; Chao, C.-Y. Highly Conductive PEDOT:PSS Films by Post-Treatment with Dimethyl Sulfoxide for ITO-Free Liquid Crystal Display. *J. Mater. Chem. C Mater.* **2015**, 3 (15), 3760–3766.

(26) Zhou, J.; Anjum, D. H.; Chen, L.; Xu, X.; Ventura, I. A.; Jiang, L.; Lubineau, G. The Temperature-Dependent Microstructure of PEDOT/PSS Films: Insights from Morphological, Mechanical and Electrical Analyses. *J. Mater. Chem. C* **2014**, 2 (46), 9903–9910.

(27) Döbbelin, M.; Marcilla, R.; Tollan, C.; Pomposo, J. A.; Sarasua, J.-R.; Mecerreyes, D. A New Approach to Hydrophobic and Water-Resistant Poly(3,4-Ethylenedioxythiophene):Poly(Styrenesulfonate) Films Using Ionic Liquids. *J. Mater. Chem.* **2008**, 18 (44), 5354.

(28) Yemata, T. A.; Zheng, Y.; Kyaw, A. K. K.; Wang, X.; Song, J.; Chin, W. S.; Xu, J. Binary Treatment of PEDOT:PSS Films with Nitric Acid and Imidazolium-Based Ionic Liquids to Improve the Thermoelectric Properties. *Mater. Adv.* **2020**, 1 (9), 3233–3242.

(29) Wu, R.; Yuan, H.; Liu, C.; Lan, J.-L.; Yang, X.; Lin, Y.-H. Flexible PANI/SWCNT Thermoelectric Films with Ultrahigh Electrical Conductivity. *RSC Adv.* **2018**, 8 (46), 26011–26019.

(30) Hynynen, J.; Järsvall, E.; Kroon, R.; Zhang, Y.; Barlow, S.; Marder, S. R.; Kemerink, M.; Lund, A.; Müller, C. Enhanced Thermoelectric Power Factor of Tensile Drawn Poly(3-Hexylthiophene). *ACS Macro Lett.* **2019**, 8 (1), 70–76.